

RANDOM FOREST CLASSIFIER

Random Forest is an ensemble method that builds multiple decision trees during training and outputs the class that is the mode of the individual trees' predictions (Breiman, 2001, cited in Witten et al., 2017). Random Forest is more robust and generalisable than a single J48 tree by averaging over 100 trees constructed from random subsets of training records and attributes. The drawback is that J48 grows only one tree that is easy to read and explain to any school governor, whereas a Random Forest of 100 trees is not as easy to visualise or explain.

I set up Random Forest in WEKA with numIterations to 100, numFeatures to 0 (which means that WEKA will use the square root of the total attributes for each split, which is the default setting) and seed to 1 to get reproducible results. I left Cross-validation at 10 folds and checked the class was (Nom) Exam_Score.

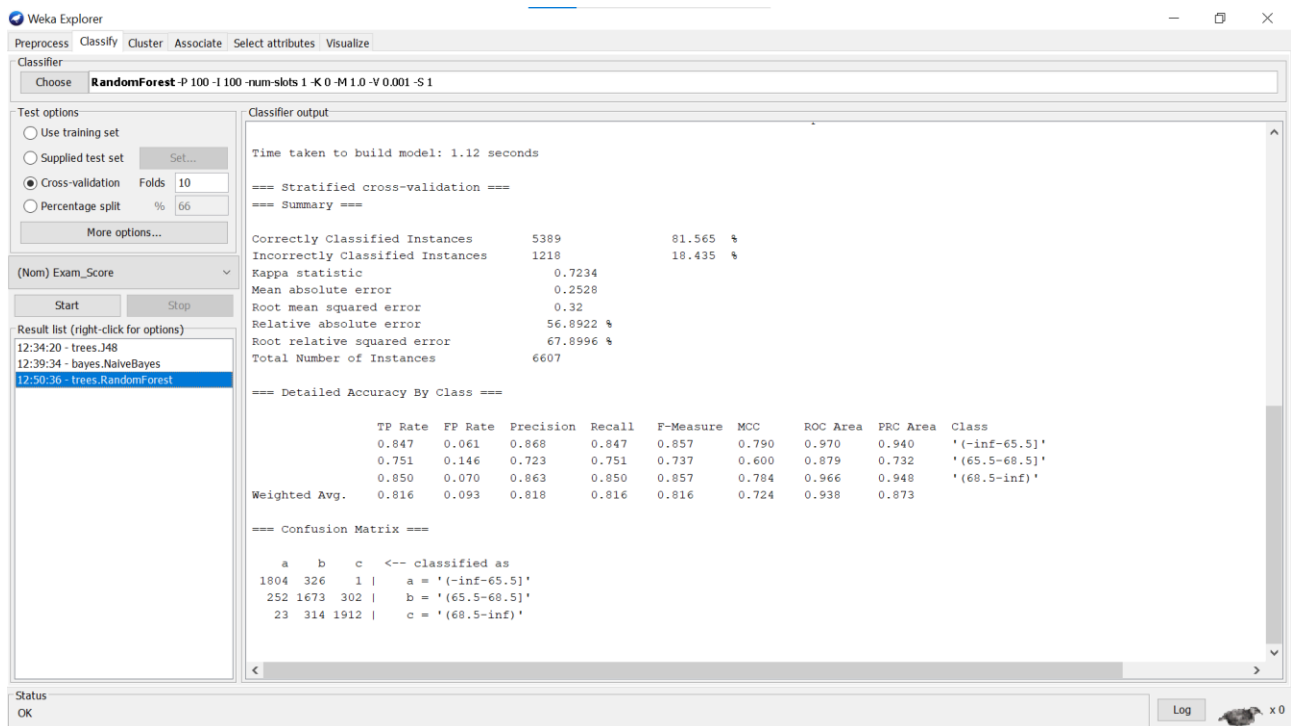


Figure 7 : WEKA Classify Tab with RandomForest -P 100 -I 100. 10-fold cross validation, class = (Nom) Exam_Score. Overall accuracy = 81.57%, Kappa statistic = 0.7234.